

Humidification

Pre-Lab Plan
Unit Operations II Lab 1
Date: 1/23/20

Commented [DL1]: 30.5/35

Group 2, Thursday, PM section

Jake Proctor
Milos Markovic
Courteney Fabbri
Erica Caruso
Mike Simon

1. Experimental Objective

The purpose of this experiment is to study the transport of both heat and mass in a wetted wall column and obtain values for heat and mass transfer coefficients. These values will be obtained through experimental methods and will be compared to values calculated using the theoretical equations. The overall goal is to qualitatively study the expected effect of system variables on the heat and mass transfer coefficients for quantitative comparison. The system variables that will serve as the independent variables for this experiment include liquid flowrate, air flowrate, and temperature of the incoming air.

Commented [DL2]: 5/5

Commented [AZ3]: Good summary of the objectives

2. Experimental

2.1 Apparatus

The apparatus setup for this experiment includes a wetted wall column, a centrifugal pump for circulating water, a heat exchanger, and various other components. The following figure (Figure 1) and table (Table 1) summarize the setup for this lab.

Commented [DL4]: 4/5

Commented [DL5]: Good. You may also include the flowsheet of the process.

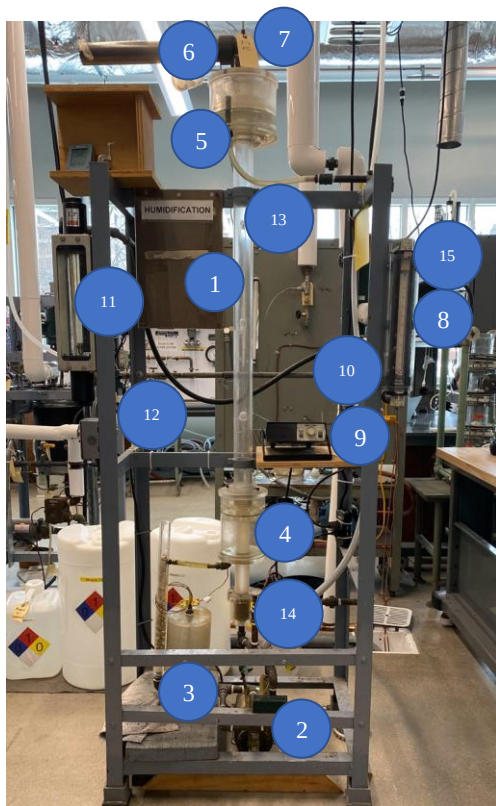


Figure 1: Apparatus setup for Humidification experiment

Table 1: Humidification experiment materials and equipment

Number	Component Name	Description
1	Column	2-in diameter plexiglass wetted column
2	Centrifugal pump	Circulates water through column
3	Water reservoir	Holds the water used for wetting the wall
4	Bottom holding tank	Used to collect water flowing through the column for recirculation
5	Top holding tank	Used to collect water at the top of the column for wetting the wall
6	Wet bulb thermometer	Measures the wet bulb temperature
7	Dry bulb thermometer	Measures the dry bulb temperature
8	Water rotameter	Measures the flowrate of water through the column
9	Digital thermometer	Reads out the temperatures measured on various channels
10	Water gate valve	Controls the amount of water flowing through the column
11	Air rotameter	Measures the flowrate of air through the column
12	Air gate valve	Controls the amount of air flowing through the column
13	Column thermometer	Measures temperature at various heights in the column
14	Heated air thermometer	Measures temperature of the air after passing through the heat exchanger
15	Heat exchanger	Uses steam to exchange heat with the compressed air

Commented [DL6]: Include the regulator.

Commented [DL7]: Unit and the maximum capacity.

Commented [DL8]: Unit.

Commented [DL9]: Unit and the maximum capacity.

Commented [DL10]: Specify the channels in the digital thermometer for all thermocouples.

2.2 Materials and Supplies

The following table summarizes the material needed for this experiment and a description of their purpose in the lab.

Table 2: Material List with Descriptions

Index	Material/Supply Name	Description
1	Water	Enters at the top of the column and flows downward countercurrent to the air
2	Low-pressure steam	Used to heat incoming air in the column; passes through the heat exchanger to heat air to different temperatures
3	Compressed air	Enters at the bottom of the column and flows upward countercurrent to water
4	Electronic hygrometer	Used to measure the amount of humidity and water vapor in the air and column outlet
5	Ladder	Used to measure the outlet humidity the at the top of the column

Commented [DL11]: 4.5/5

Commented [DL12]: Ambient temperature and humidity digital meter is needed.

2.3 Experimental Plan

Commented [DL13]: 8/10

Prelab Procedures:

1. Measure the humidity of the incoming air by using the wet and dry bulb thermometers (or the electronic hygrometer)
2. Measure the air temperature

Lab Procedures:

1. Circulate water in the column at a sufficient rate to completely wet the inside wall of the tube
2. Establish an air flow (unheated) and allow the circulating water temperature to reach steady state
3. Measure the steady state values of the water and air temperature at various locations in the column as well as the air humidity using the wet and dry bulb thermometers (or electronic hygrometer)
4. Calculate the values of k_y , h and N_{IG} using equations X and Y in the lab manual
5. Repeat above for two other air flowrates
6. Repeat above for the same three air flowrate, but with heated air

Commented [DL14]: How much?

Commented [DL15]: What's the criteria of the steady state?

Commented [DL16]: Be specific.

Commented [DL17]: Be specific.

Project Safety Evaluation

Commented [DL18]: 9/10

Project Title: Humidification

Date: 1/23/20

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: Electrical outlet near water – use caution
Potential mechanical hazards?	Yes	If yes, identify conditions: Centrifugal pump pumping water can cause injury,
Potential pressure hazards?	Yes	If yes, identify conditions: Compressed air flowing through system – ensure regulator is properly set for air flow
Potential temperature hazards?	Yes	If yes, identify conditions: Wetted wall column will be hot when in operation – use caution, also heat exchanger gets hot
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	No	If yes, specify type:
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Water and Air (FIND NUMBER)
Process Parameters	Max Expected	List location and possible hazards and precautions
Temperature (°C)	100	Wetted wall column and heat exchanger (and hot water)
Pressure (psig)	50	Compressed air should be regulated to 50 psig max
Safety Controls	yes/no	If yes, Provide description
	yes	Correct regulator setting for air flow and setting on heat exchanger is important for safe operation

Commented [DL19]: Yes.

Commented [DL20]: Steam is at 100-120C. Your air temperature won't be as high as this.

Commented [DL21]: Also give the max pressure of the compressed air before the regulator

Commented [AZ22]: Shutoff valve for the compressed air and valve for the steam, pump switch.

Commented [DL23]: The flow rate and temperature are controlled.
20-amp surge protector.
A regulator is attached to the system to allow complete control of air to prevent high pressure build up.

I have read relevant background material for the Unit Operations Laboratory entitled: “Humidification” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

<u>Jake Proctor</u>	<u>1/23/2020</u>
signature of team member	date
<u>Courteney Fabbri</u>	<u>1/23/2020</u>
signature of team member	date
<u>Milos Markovic</u>	<u>1/23/2020</u>
signature of team member	date
<u>Erica Caruso</u>	<u>1/23/2020</u>
signature of team member	date
<u>Mike Simon</u>	<u>1/23/2020</u>
signature of team member	date